

HYDROGENATED NBR: PATENT LANDSCAPE AND SYNTHESIS-RELEVANT EVIDENCE FROM THE PROVIDED RECORD

1. ABSTRACT

The provided patent record does not contain direct hydrogenated NBR synthesis disclosures in either of the two primary patents. Instead, both patents are technically adjacent examples of hydrogenation-enabled polymer/material modification: US10927052B2 describes oligomerization of branched C10 olefins followed by optional hydrogenation of the resulting olefin oligomers, while US11958974B2 describes asphalt rejuvenation using a partially hydrogenated bio-oil blended with a non-hydrogenated bio-oil. For the target material hydrogenated NBR (HNBR), these patents are therefore only indirectly informative, mainly at the level of hydrogenation handling, process control, and post-reaction property tuning rather than nitrile-butadiene rubber chemistry.

US10927052B2 emphasizes feed pretreatment, Lewis-acid/ionic-liquid/clay catalysis, and downstream hydrogenation of oligomer fractions. The disclosed process includes silica-gel treatment of mixed olefins, oligomerization using AlBr_3 /isomerized decenes at about 100 °C over 3 h, and a BF_3 /n-butanol route at 75 °C and 20 psig for 2 h, followed by neutralization, washing, drying, distillation, and hydrogenation of C20+ fractions. The patent highlights control of feed purity, catalyst choice, and product fractionation before hydrogenation.

US11958974B2 focuses on blending a partially hydrogenated bio-oil with a non-hydrogenated bio-oil to tune asphalt rheology. The examples report blending at 155 °C $\pm$ 10 °C, 3000 rpm, with dosage combinations from 0.5% to 10%, and show that combined partially hydrogenated and non-hydrogenated materials reduce stiffness and viscosity while improving low-temperature performance. The work uses DSR, BBR, rotational viscosity, and SCB testing to optimize dosage and performance grade.

Across the two patents, the recurring themes relevant to HNBR are selective hydrogenation, property-driven partial saturation, and process/property optimization through controlled feed composition, temperature management, and post-treatment characterization. However, no patent in the required manifest explicitly discloses HNBR polymerization, NBR hydrogenation conditions, residual unsaturation targets, or HNBR-specific catalysts.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US10927052B2
- Patent Title: Decene oligomers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example 3: - Example Extracted Example Example 4: - Example Extracted Example Example 3: - Example Extracted Example Example 4.: - Example Extracted Example Example 5: - Example Extracted Example Example 5: - Example Extracted Example Example 6:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 2

- Patent Number: US11958974B2
- Patent Title: Rejuvenation of vacuum tower bottoms through bio-derived materials
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2024
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example 3: - Example Extracted Example Example 4: - Example Extracted Example Example 5: - Example Extracted Example Example 6: - Example Extracted Example Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example 3: - Example Extracted Example Example 7: - Example Extracted Example Example 8: - Example Extracted Example Example 9: - Example Extracted Example Example 10: - Example Extracted Example Example 11: - Example Extracted Example Example

12: - Example Extracted Example Example 13: - Example Extracted Example Example 14: - Example Extracted Example Example 15: - Example Extracted Example Example 16: - Example Extracted Example Example 17:

Technical Relevance Selected via deterministic pipeline scoring.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- Both primary patents are hydrogenation-adjacent but not HNBR-specific; neither discloses nitrile-butadiene rubber monomer composition, NBR polymerization, or HNBR residual unsaturation targets.
- US10927052B2 is a hydrocarbon oligomerization/hydrogenation process: branched C10 olefin feedstocks, Lewis-acid and ionic-liquid catalyst systems, and downstream hydrogenation of oligomer fractions. US11958974B2 is a formulation/blending patent using partially hydrogenated bio-oil in asphalt, not a polymer hydrogenation route.
- US10927052B2 reports explicit reaction conditions for oligomerization: silica-gel pretreatment, AlBr₃/isomerized decenes at about 100 °C for 3 h, and BF₃/n-butanol at 75 °C and 20 psig for 2 h. US11958974B2 reports blending conditions: 155 °C ± 10 °C, 3000 rpm, 1 h blending, with 0.5% to 10% dosage ranges.
- US10927052B2 uses catalyst systems and feed purification to enable oligomer formation before hydrogenation; US11958974B2 uses a partially hydrogenated component as a rheology modifier and does not describe catalytic hydrogenation in the examples provided.
- US10927052B2 includes neutralization/washing steps after oligomerization (basic water, sodium hydroxide wash, water washes, filtration, drying, distillation) before hydrogenation. US11958974B2 emphasizes aging/testing workflows (RTFO, PAV, DSR, BBR, SCB) rather than chemical workup.
- Historically, US10927052B2 reflects a more classical process-chemistry approach with catalyst selection and product fractionation; US11958974B2 reflects a formulation/property-optimization approach using partially hydrogenated feedstocks and statistical modeling.
- Neither patent provides a direct analogue for HNBR hydrogenation selectivity, but both support the broader principle that partial hydrogenation and controlled process conditions are used to tune final material performance.

4. CONCLUSION

Based on the provided evidence, the required patent set does not directly disclose hydrogenated NBR synthesis or manufacturing. The most relevant transferable lessons are process discipline around hydrogenation-adjacent transformations: feed pretreatment to remove polar impurities, careful temperature control, staged addition of catalyst or modifier, neutralization/washing, fractionation before hydrogenation, and property verification after modification. For actual HNBR

synthesis, the evidence here is insufficient to recommend specific NBR hydrogenation catalysts, pressures, or residual unsaturation targets. The best-supported conclusion is that HNBR development should be sought in patents explicitly directed to NBR hydrogenation rather than in these adjacent hydrocarbon oligomer and asphalt-modification disclosures. If one were to extrapolate only from the present record, the most defensible synthesis principle is controlled partial hydrogenation under tightly managed thermal conditions, followed by rigorous property and composition analysis.

5. REFERENCES

- US10927052B2 | Decene oligomers | Not disclosed | US | 2021 | <https://patents.google.com/patent/US10927052B2/en>
- US11958974B2 | Rejuvenation of vacuum tower bottoms through bio-derived materials | Not disclosed | US | 2024 | <https://patents.google.com/patent/US11958974B2/en>